

SmartHire AI — System Architecture (v1.0)

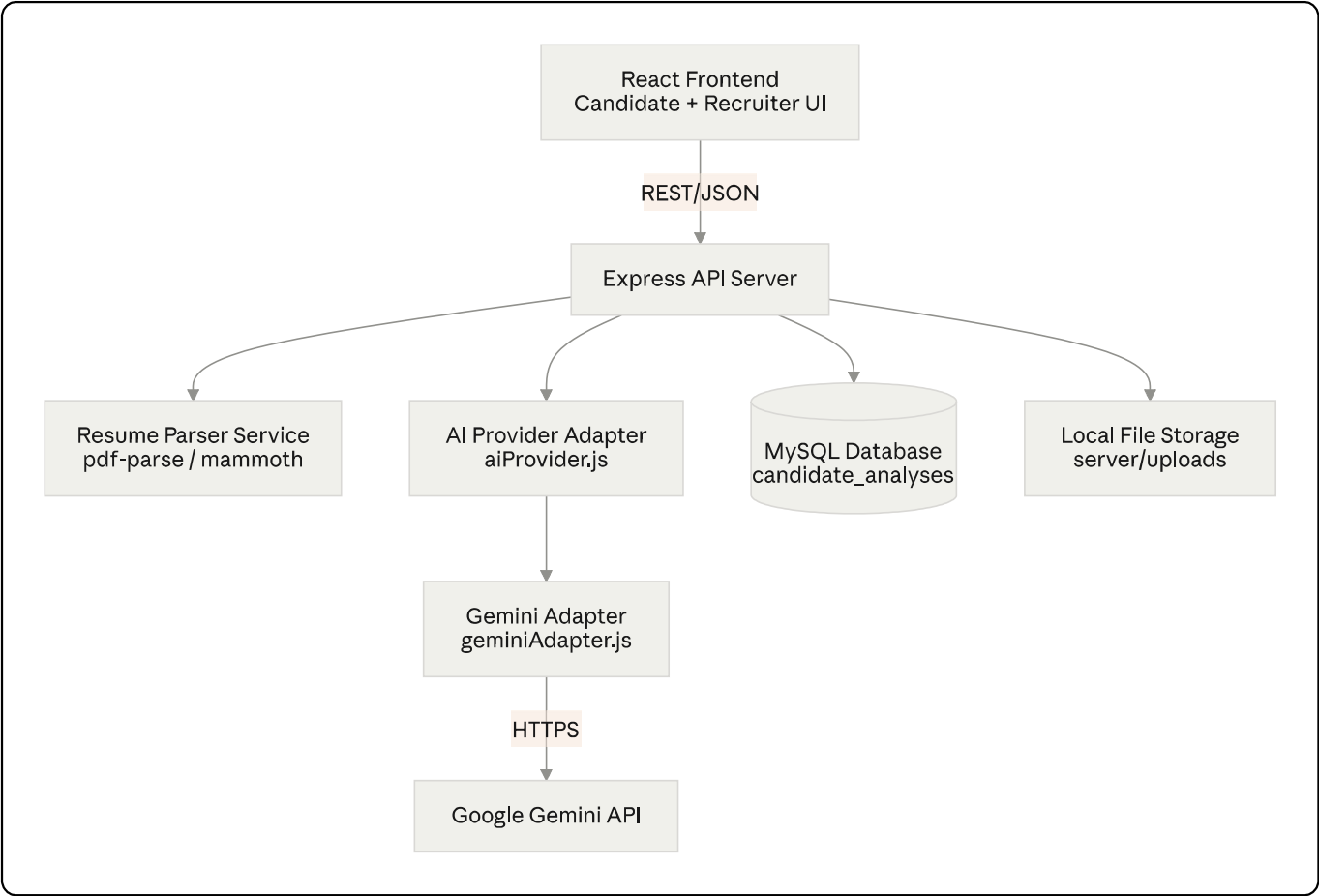
Source of truth: PRD + Implementation Blueprint (Day 1). No redesign — this document adds technical detail only.

1. Tech Stack (confirmed, no changes)

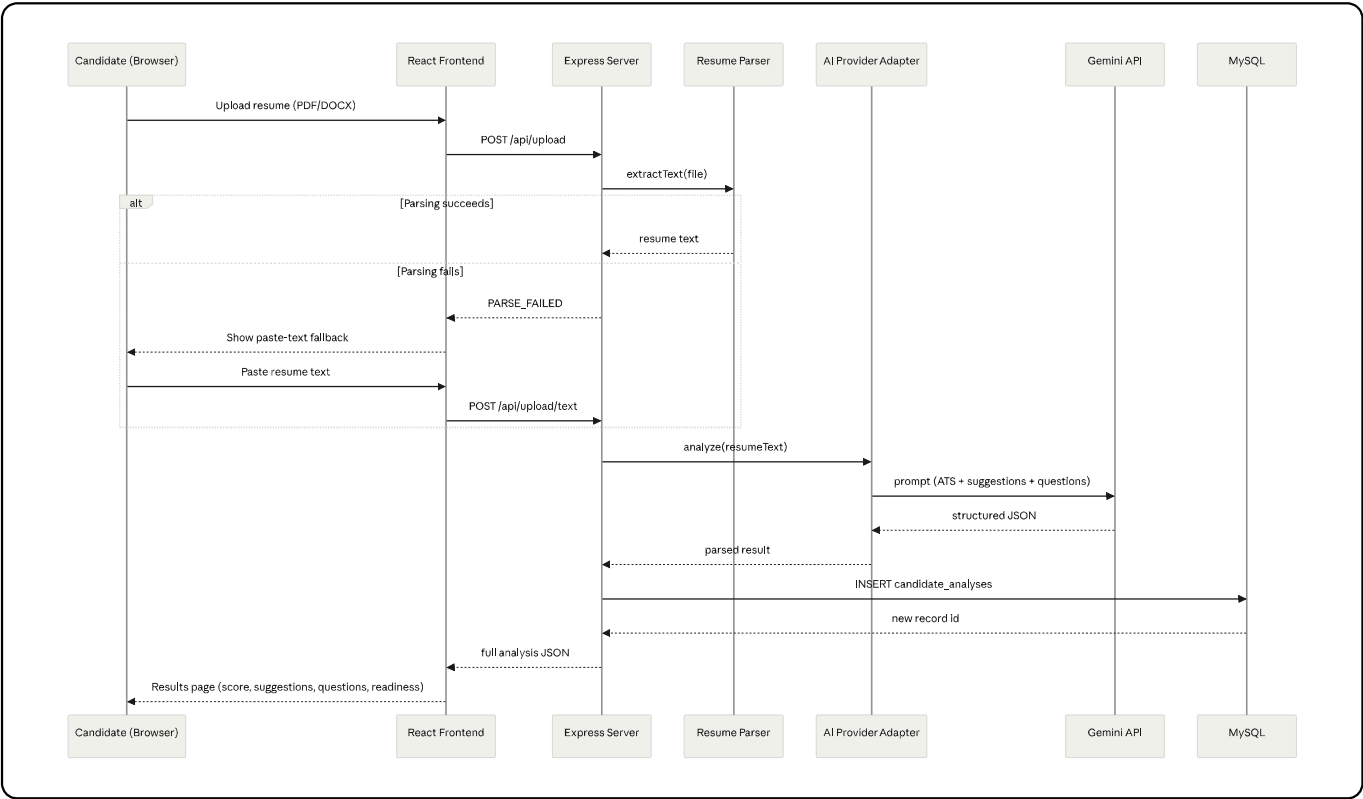
Layer	Choice	Why
Frontend	React (Vite)	Fast dev server, component-driven UI, matches your existing skills
Backend	Node.js + Express	Lightweight REST API, easy file upload handling, matches your skills
Database	MySQL	Relational, simple schema, matches your skills, AWS RDS free tier available
Auth	None (v1.0)	Explicitly out of scope per PRD — keeps scope lean
AI	Google Gemini API via swappable adapter	Free tier, generous limits, adapter pattern allows swapping to Claude/OpenAI later
Hosting	AWS EC2 + RDS (Free Tier)	Matches your AWS learning goal, real cloud deployment
File Parsing	pdf-parse, mammoth	Free, npm-based, no external API cost
Process Manager	pm2	Keeps Node server alive in production

No paid tools used anywhere in v1.0.

2. Component Diagram



3. Data Flow (Candidate Analysis)



4. Request Lifecycle (typical API call)

1. Frontend sends request with JSON or multipart/form-data body.
2. Express middleware: CORS check → body parsing / Multer (for file routes) → route handler.
3. Controller validates input (file type/size, required fields).
4. Controller calls the relevant service (parser, AI adapter, or DB layer).
5. Response is shaped into a consistent JSON envelope and returned.
6. Errors are caught by a centralized error handler and returned as `{ error: { code, message } }`.

5. AI Interaction Design

- All AI calls go through `aiProvider.js` — the rest of the app never calls Gemini directly.
- `aiProvider.js` exposes: `generateATSAnalysis()`, `generateSuggestions()`, `generateInterviewQuestions()`.
- `geminiAdapter.js` implements these against the Gemini API with strict JSON-only prompts.
- To swap providers later (Claude/OpenAI), only a new adapter file is added — no changes needed to routes, controllers, or the database layer.
- A JSON-safety layer strips markdown fences and retries once on malformed output before failing gracefully.

6. External Services

Service	Purpose	Tier
Google AI Studio (Gemini API)	Resume/AI analysis	Free tier
AWS EC2	App hosting (frontend + backend)	Free tier (t2.micro/t3.micro)
AWS RDS (or EC2-hosted MySQL fallback)	Database	Free tier
GitHub	Source control	Free

7. Key Architectural Decisions

- **No authentication in v1.0** — confirmed from PRD; all data is public by design for this version.

- **Single analysis table, no joins** — keeps schema and queries simple, matches PRD data model exactly.
- **Provider-agnostic AI layer** — the single most important architectural decision protecting future flexibility.
- **Graceful degradation** — every external dependency (parser, AI, DB) has a defined failure path so the app never shows a blank/broken screen.